Machine Learning

Lecture 9: Metrics, Thresholds, and Asymmetric Costs

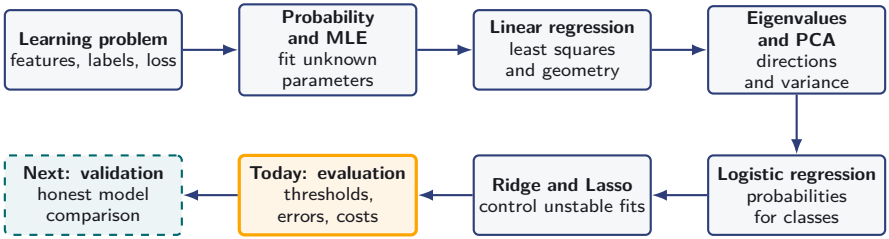
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Civil Engineering; Computer Science and Engineering

Semester I, 2026



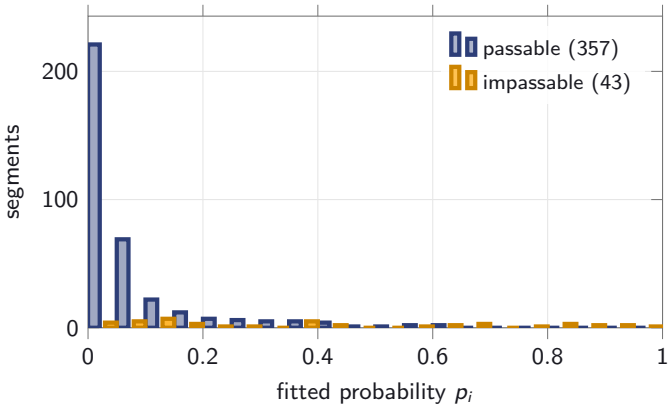
Course Flowchart: What Each Step Added



Summary

The earlier lectures produced fitted probabilities. Today we learn how to turn those probabilities into decisions and how to judge the consequences.

Where The Probability Model Left Us



The road-segment model gives every segment a probability of becoming impassable,

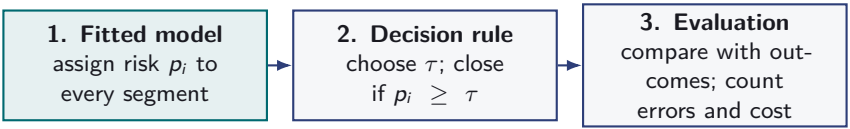
$$p_i = \sigma(\theta_0 + \theta_1 \,\mathrm{rain}_i + \theta_2 \,\mathrm{wet}_i),$$

fitted by maximum likelihood on 400 segments.

Question

The two histograms overlap. Whatever line we draw, some impassable segments fall on the "open" side and some passable ones on the "close" side. **How do we choose the line, and how do we report how good it is?**

The Procedure This Lecture Fills In



Two different objects.

- p_i is a *probability*; the model produced it.
- $\widehat{y}_i = \mathbb{I}[p_i \geq \tau]$ is a *decision*; we produce it.

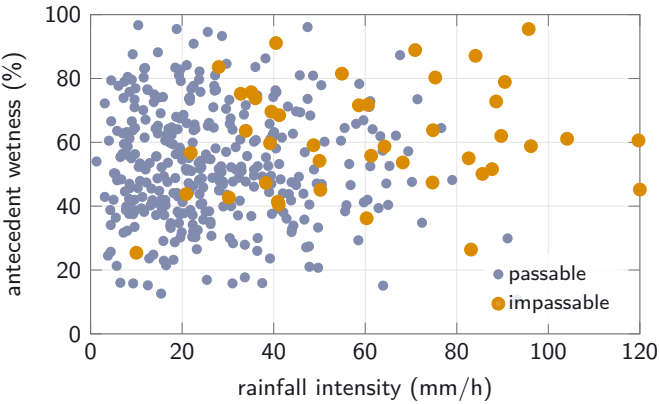
Three questions, in order.

1. What happens at one threshold? (confusion matrix)
2. What happens across all thresholds? (ROC, PR)
3. Which threshold should we use? (costs)

Summary

The threshold is not part of the model. Refitting is never required to change it.

The Dataset: 400 Road Segments



Each segment has two features and one outcome from the last monsoon:

- rainfall intensity (mm/h), antecedent wetness (%);
- $y_i = 1$ if the segment became impassable.

Class balance: 43 impassable, 357 passable, so the prevalence is

$$\pi = \frac{43}{400} = 0.1075.$$

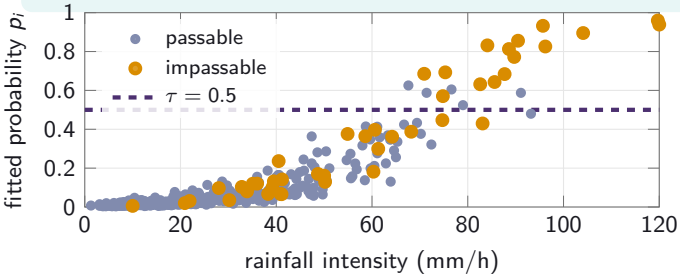
Warning

Keep the number 0.1075 in view. Half of today's warnings are about what a rare positive class does to a metric.

Road-Closure Decision Problem

Complete problem

Given fitted risks p_i for 400 road segments, choose a threshold τ, close when $p_i \geq \tau$, and count missed impassable roads and unnecessary closures.



What one dot tells us

- Horizontal position: rainfall.
- Vertical position: fitted risk p_i; the model already used rain and wetness.
- Colour: actual outcome; orange means impassable.

Question

At $\tau = 0.5$, mark $p_i \geq \tau$ as "close." Compare each decision with the actual colour. Count the misses and unnecessary closures.

The Story: The Night Before The Storm

A district road controller has 400 segments on her map and a forecast of heavy rain by morning. From the last monsoon she has, for every segment, the rainfall it saw, how wet the ground was, and whether it became impassable. The logistic model from the previous lectures turns tonight's forecast into a number p_i per segment: the chance it floods.

She has one crew and one decision per segment: **close it, or leave it open**. Closing a road that stays passable costs diversions and complaints, about one unit. Leaving a road open that floods strands people and needs a rescue: about five units.

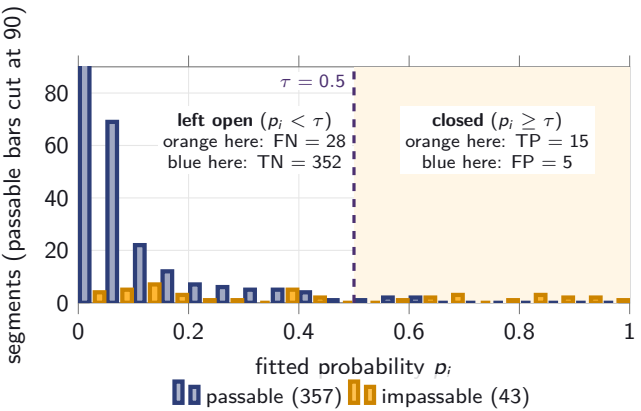
Question

Four people will question her tomorrow.

- **Her supervisor:** how many did you get right?
- **The finance officer:** how many closures were needless?
- **The engineer:** is the model any good, whatever line you drew?
- **The auditor:** does a 0.3 really mean three in ten?

Each question is a different metric. The lecture answers them in that order, then prices the decision.

Where The Four Counts Come From: A Line On The Histogram



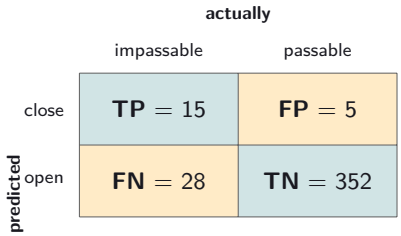
The model gave every segment a number p_i. A threshold is nothing more than a **vertical line** through this picture: everything to its right is closed, everything to its left is left open. Orange bars to the right: **TP**. Blue to the right: **FP**. Orange to the left: **FN**. Blue to the left: **TN**.

Insight

Move the line and the four counts change; the model does not. That is the mechanism behind every curve today.

Interactive explorer: drag the line and watch the four counts ↗

One Threshold, Four Outcomes



$\tau = 0.5$; rows are our decision, columns are what happened.

At $\tau = 0.5$:

- **TP** $= 15$: closed; it did become impassable.
- **FP** $= 5$: closed; it stayed passable (*false alarm*).
- **FN** $= 28$: left open; it became impassable (*miss*).
- **TN** $= 352$: left open; it stayed passable.

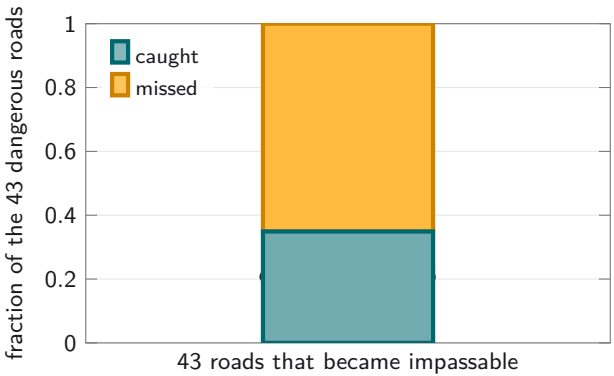
Confusion matrix. This 2×2 table records the four outcomes at one threshold. Every scalar metric uses these counts.

Warning

At $\tau = 0.5$ we miss 28 of 43 impassable segments. Lowering τ trades some misses for additional false alarms.

Interactive explorer: drag the threshold and watch the four cells move ↗

Question 1: Of The Dangerous Roads, How Many Did We Catch?



First ignore all 357 passable roads. Keep only the 43 roads that actually became impassable.

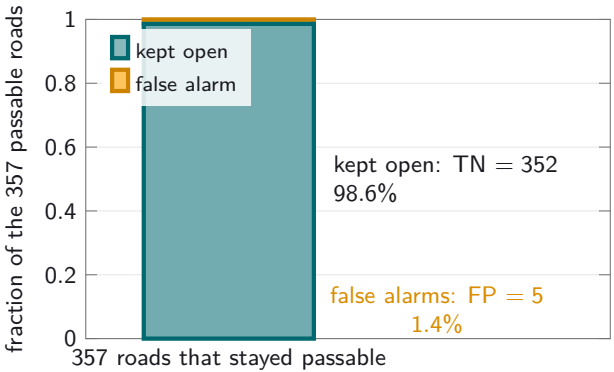
$$\underbrace{15}_{\text{caught}} + \underbrace{28}_{\text{missed}} = 43.$$

Recall or true-positive rate

$$\text{recall} = \frac{15}{43} = 0.349.$$

The rule caught about 35% of the roads that became impassable. The remaining 65% were missed.

Question 2: Of The Passable Roads, How Many Did We Alarm?



Now ignore the dangerous roads. Keep only the 357 roads that actually stayed passable.

$$\underbrace{352}_{\text{kept open}} + \underbrace{5}_{\text{false alarms}} = 357.$$

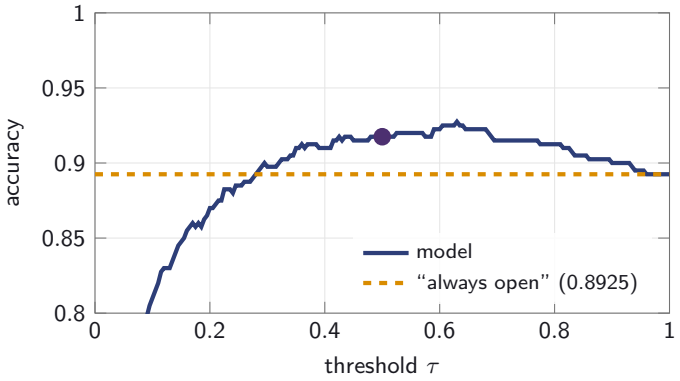
False-positive rate

$$\mathrm{FPR} = \frac{5}{357} = 0.014.$$

Only about 1.4% of passable roads were closed unnecessarily. ROC uses this rate on its horizontal axis and recall on its vertical axis.

Accuracy, And Why It Lies Here

The story: The supervisor asks how many she got right. Accuracy answers, and the answer flatters her.



Accuracy at $\tau = 0.5$:

$$\frac{\mathrm{TP} + \mathrm{TN}}{n} = \frac{15 + 352}{400} = 0.9175.$$

A rule that never closes any road, $\widehat{y}_i \equiv 0$, scores

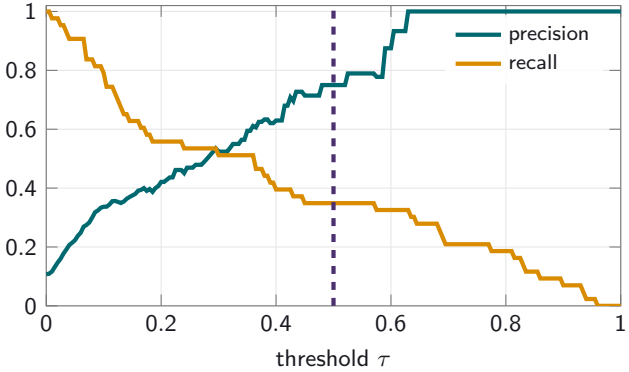
$$\frac{357}{400} = 0.8925.$$

Warning

The model beats "do nothing" by 2.5 points while missing 65% of the dangerous segments. Accuracy weights the classes by their size, and here one class is nine times the other.

Precision And Recall Look At The Same Cell Two Ways

The story: The finance officer asks how many closures were needless; the supervisor asks how many floods she caught. Two questions, two metrics.



Both start from TP; they divide by different totals.

$$\text{precision} = \frac{\text{TP}}{\text{TP+FP}} = \frac{15}{20} = 0.750,$$

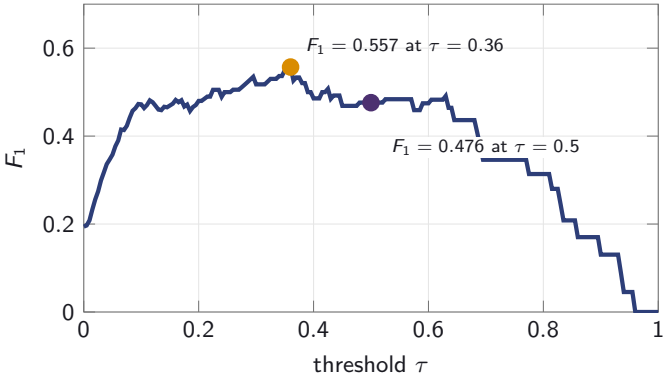
$$\text{recall} = \frac{\text{TP}}{\text{TP+FN}} = \frac{15}{43} = 0.349.$$

Precision: of the roads we closed, how many needed it. Recall: of the roads that needed it, how many we closed. Lowering τ closes more roads, so recall can only rise while precision usually falls; they cross near $\tau \approx 0.36$.

Warning

Precision is undefined when nothing is closed; software reports 1 there. The right edge of the teal curve is convention, not evidence.

F_1: One Number From Two, And What It Hides



The harmonic mean of precision and recall,

$$F_1 = \frac{2\,\mathrm{P}\,\mathrm{R}}{\mathrm{P}+\mathrm{R}} = \frac{2(0.750)(0.349)}{0.750+0.349} = 0.476 \quad (\tau = 0.5).$$

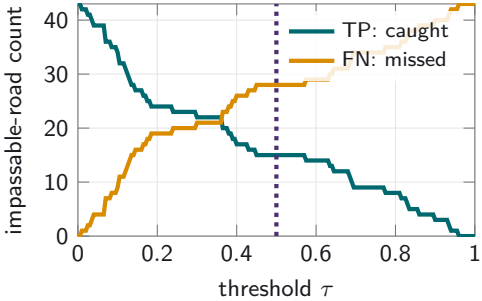
The harmonic mean is dragged toward the *smaller* of the two: a recall of 0.349 caps F_1 below 0.52 no matter how good precision is.

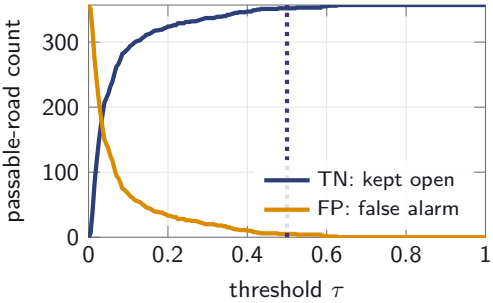
Warning

F_1 weighs a false alarm and a miss equally. Nothing about a road closure says they are equal. Use F_1 to compare models when you have no cost information; do not use it to *choose* a threshold when you do.

Every Cell Moves When τ Moves

Problem. Use a different threshold on the same fitted probabilities. Which decisions change, and which totals must remain fixed?





Lower τ: TP and FP rise, FN and TN fall. The totals do not move: $\mathrm{TP} + \mathrm{FN} = 43$ and $\mathrm{FP} + \mathrm{TN} = 357$. One threshold gives one confusion matrix.

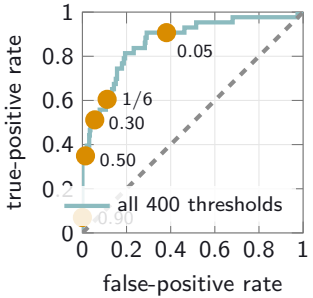
Interactive explorer: sweep τ and watch the counts trade ↗

How A Curve Is Built: One Row Per Threshold

Pick a few thresholds, read the four counts at each, compute the rates: each row is one point.

τ	TP	FP	FN	TN	TPR	FPR	prec.
0.90	3	0	40	357	0.07	0.00	1.00
0.50	15	5	28	352	0.35	0.01	0.75
0.30	22	20	21	337	0.51	0.06	0.52
1/6	26	40	17	317	0.60	0.11	0.39
0.05	39	136	4	221	0.91	0.38	0.22

$\mathrm{TPR} = \mathrm{TP}/43$, $\mathrm{FPR} = \mathrm{FP}/357$, precision $= \mathrm{TP}/(\mathrm{TP}+\mathrm{FP})$. Row $\tau = 0.3$: we close 42 roads, 22 rightly: recall 22/43, false alarms 20/357, precision 22/42.

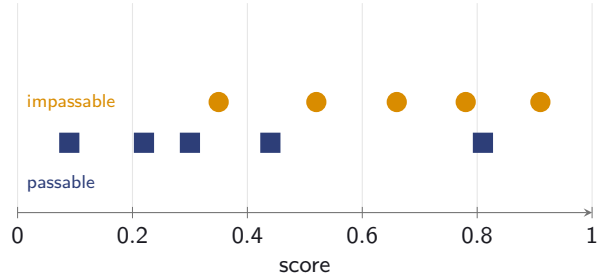


Summary

Plot (FPR, TPR) from each row: the orange points. Every possible τ joins them into the ROC curve; plotting (recall, precision) from the same rows gives the PR curve.

ROC By Hand: Ten Scores

The story: The engineer asks whether the model is any good regardless of where she drew the line. To answer, take the line away and look at every line at once.



score	.91	.81	.78	.66	.52	.44	.35	.30	.22	.09
class	+	−	+	+	+	−	+	−	−	−

Problem. Build an ROC curve directly from these ten labelled scores. Five are impassable and five are passable. Slide τ from above 0.91 down to below 0.09 and record $(\mathrm{FPR}, \mathrm{TPR})$ after each score is passed:

- pass a $+$: TPR rises by $1/5$ (step *up*);
- pass a $-$: FPR rises by $1/5$ (step *right*).

The sequence $+\ -\ +\ +\ +\ -\ +\ -\ -\ -$ gives the path

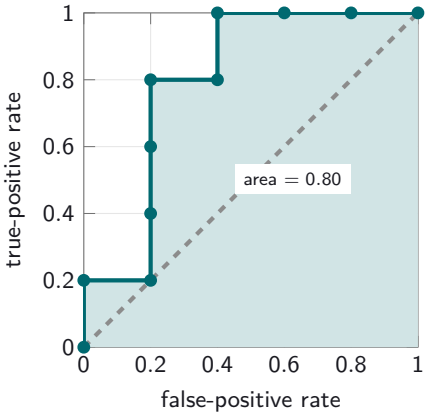
$$\uparrow \rightarrow \uparrow\uparrow\uparrow \rightarrow \uparrow \rightarrow\rightarrow\rightarrow .$$

ROC curve

The set of $(\mathrm{FPR}, \mathrm{TPR})$ pairs over all thresholds. For finite data it is a staircase from $(0, 0)$ to $(1, 1)$.

The Staircase, And Its Area

In one sentence: The ROC curve is drawn by lowering the threshold one score at a time; the area under it counts how often an impassable road outscores a passable one.



Column by column (FPR width 0.2 each):

$$0.2\,(0.2 + 0.8 + 1 + 1 + 1) = 0.80.$$

Now count *pairs*. Of the $5 \times 5 = 25$ (impassable, passable) pairs, how many has the model ordered correctly, impassable scored higher?

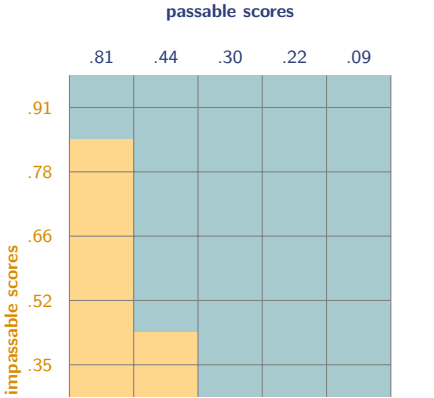
$+\backslash-$	.81	.44	.30	.22	.09
.91	✓	✓	✓	✓	✓
.78	×	✓	✓	✓	✓
.66	×	✓	✓	✓	✓
.52	×	✓	✓	✓	✓
.35	×	×	✓	✓	✓

20 of 25 correct: $20/25 = 0.80$. **The same number.**

Interactive explorer: count the pairs yourself, then check the area ↗

AUC Is A Probability About Pairs

In one sentence: AUC $= 0.867$ means: pick one impassable and one passable road at random, and the model ranks them correctly 87% of the time. It says nothing about which threshold to use.



teal: correctly ordered (20); orange: reversed (5)

Area under the ROC curve

$$\mathrm{AUC} = \Pr\left(s_+ > s_-\right),$$

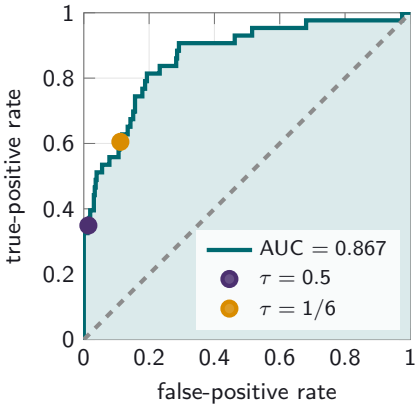
the probability that a random positive is scored above a random negative (ties count $\frac{1}{2}$).

- 0.5: random ordering (the diagonal); 1: every positive above every negative.
- It never mentions τ: AUC measures the *ranking*, not a decision.

For all 400 segments. Pair counting and trapezoid area both give $\mathrm{AUC} = 0.8671$.

The ROC Curve For The 400 Segments

In one sentence: Every threshold is one point on this curve; moving τ slides along it, and only a better model can lift it.



Each point is one threshold; the curve is every threshold at once.

- $\tau = 0.5$: $(5/357,\ 15/43) = (0.014,\ 0.349)$: cautious about closing, so it misses.
- $\tau = 1/6$: $(40/357,\ 26/43) = (0.112,\ 0.605)$, further up the curve.

Interpretation. Moving τ slides along the curve; changing the fitted model changes the curve itself.

Question

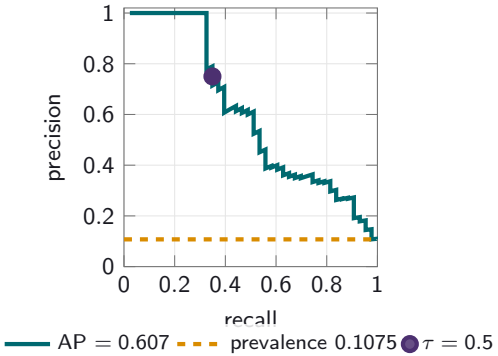
Where should we operate? The curve does not know what a miss costs.

Interactive explorer: move τ and watch the operating point slide along the curve ↗

Precision–Recall: The Curve That Feels The Prevalence

The story: The finance officer: forty needless closures out of 357 passable roads sounds small; out of 66 closures it does not.

In one sentence: Precision asks "of the roads we closed, how many needed it?", so it feels the rare class where the ROC does not.



Problem. Can 40 false alarms look small among 357 passable roads but large among the roads we close? Plot recall against precision.

- The floor is the prevalence, 0.1075: closing every road has precision 43/400.
- Area under this curve (average precision) is $\mathrm{AP} = 0.607$.
- The curve is jagged at low recall: a single false alarm among the first few closures moves precision a lot.

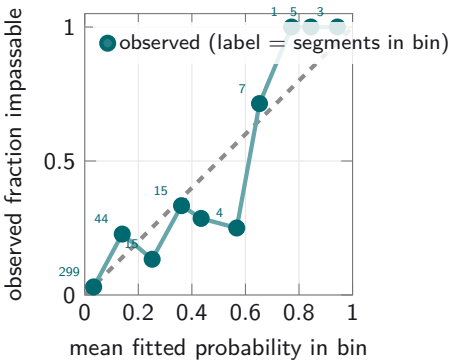
Why PR matters. ROC gives $40/357 = 0.11$; precision gives $26/66 = 0.39$. The latter exposes the burden among roads actually closed.

Interactive explorer: the same threshold on both curves ↗

Calibration: Does $p = 0.3$ Mean Thirty Percent?

The story: The auditor asks whether the model's numbers can be trusted as probabilities, because the pricing that follows depends on it.

In one sentence: Calibration checks whether the model's 0.3 really means three in ten; the cost threshold only works if it does.



Bin the segments by fitted p_i (width 0.1); compare each bin's mean p with the fraction that flooded.

- Bin $[0, 0.1)$: mean $p = 0.033$, observed 0.030, $n = 299$. On the line.
- Bin $[0.1, 0.2)$: mean $p = 0.140$, observed 0.227, $n = 44$. *Above* the line: riskier than the model says.
- Bins above 0.5 hold 20 segments in total; their points are noise.

Interactive explorer: rebin and watch the points settle or scatter ↗

One Road, Two Possible Decisions

The story: Each road is a small bet; the controller takes the cheaper side.

In one sentence: Compare the risk of closing, $c_{\mathrm{FP}}(1-p)$, with the risk of leaving open, $c_{\mathrm{FN}}\,p$; take the smaller.

One road, $p = 0.20$; prices $c_{\mathrm{FP}} = 1$, $c_{\mathrm{FN}} = 5$.

Decision A: close the road

Needless only if the road stays passable, probability $1 - p = 0.80$:

$$\text{expected cost} = 0.80 \times 1 = \boxed{0.80}.$$

Decision B: leave it open

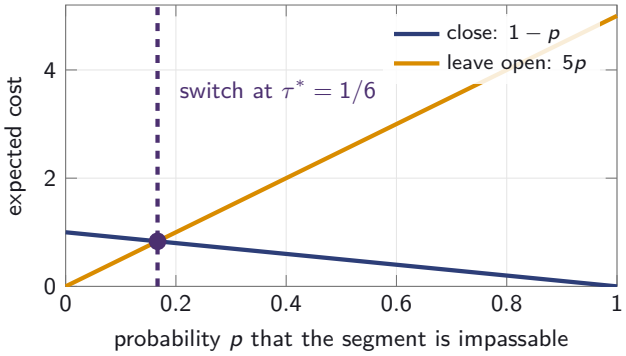
Costly only if the road floods, probability $p = 0.20$:

$$\text{expected cost} = 0.20 \times 5 = \boxed{1.00}.$$

Choose "close": $0.80 < 1.00$. A 20% risk is enough because a miss costs five times a false alarm.

From One Road To The Cost-Based Threshold

In one sentence: The two risks are equal at $p = c_{\mathrm{FP}}/(c_{\mathrm{FP}} + c_{\mathrm{FN}})$; that number is the threshold, and nothing else in the model changes.



For any p: close costs $1 - p$, leave open costs $5p$. The decision changes where they are equal:

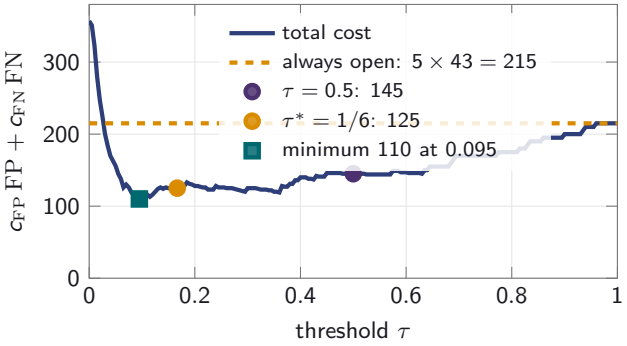
$$1 - p = 5p \quad \Longrightarrow \quad 1 = 6p \quad \Longrightarrow \quad p = \frac{1}{6}.$$

Decision rule

Leave open below $1/6$; close above $1/6$. At exactly $1/6$, both decisions have the same expected cost.

Total Cost Across Thresholds

In one sentence: Add up the prices of all false alarms and misses at each τ: the curve's bottom is the cheapest threshold on this data.



Sum the prices over the confusion matrix, $\mathrm{cost}(\tau) = c_{\mathrm{FP}}\mathrm{FP}(\tau) + c_{\mathrm{FN}}\mathrm{FN}(\tau)$:

τ	FP	FN	cost
0.5	5	28	$5 + 140 = 145$
1/6	40	17	$40 + 85 = 125$
0.095	60	10	$60 + 50 = 110$

From 0.5 to τ^*: 14% cheaper with the *same model*.

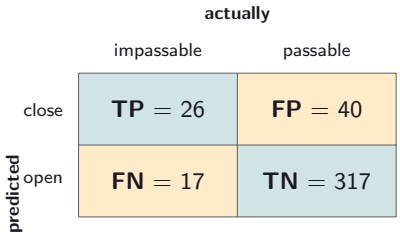
Question

Why is the minimum at 0.095, not at 1/6? The model slightly under-states risk near $p \approx 0.14$, and 43 positives make the curve noisy (appendix).

Interactive explorer: set your own prices and read off τ^* ↗

The Confusion Matrix At τ^*

In one sentence: Moving from 0.5 to 1/6 makes accuracy and precision worse and the decision better; report the cost, not the headline metrics.



$\tau = 1/6$

	$\tau = 0.5$	$\tau = 1/6$
accuracy	0.918	0.858
precision	0.750	0.394
recall	0.349	0.605
F_1	0.476	0.477
cost	**145**	**125**

Insight

Accuracy fell six points, precision halved, F_1 did not move, and the decision got *better*. Report only the first three rows and you would reject the change.

Three Ways To Choose τ, And What Each Assumes

In one sentence: Default 0.5 is a hidden assumption that both errors cost the same; with agreed prices, use τ^*.

rule	what it assumes	on our data	total cost
default $\tau = 0.5$	a miss and a false alarm cost the same; nobody checked	$(15, 5, 28, 352)$	145
best F_1	no price information; balance precision and recall	$\tau = 0.36$	≈ 135
price ratio $\tau^* = c_{\mathrm{FP}}/(c_{\mathrm{FP}} + c_{\mathrm{FN}})$	prices agreed; p calibrated	1/6: $(26, 40, 17, 317)$	125
minimum of the cost curve	the training set is the world	$\tau = 0.095$	110 (optimistic)

Summary

Recipe: agree the prices; compute τ^*; check calibration near it; report the confusion matrix *at that threshold* and the cost.

Warning

The training-set minimum is tempting; its extra saving over τ^* is partly noise from 43 positives and will not all survive on new segments. Next lecture.

Which Metric, When

In one sentence: Pick the metric that answers the question the decision-maker will actually ask.

question	metric	fails when
Does the model rank positives above negatives?	AUC	you already know where you will operate
How good is one operating point?	confusion matrix and its rates	reported without the threshold that produced it
How many closures were wasted?	precision	the class is balanced (then FPR says the same thing)
How many dangers were caught?	recall / TPR	reported alone (recall 1 is free: close everything)
Can I act on p as a frequency?	reliability diagram	fewer than a few dozen positives per bin
What does the decision cost?	$c_{\mathrm{FP}}\mathrm{FP} + c_{\mathrm{FN}}\mathrm{FN}$	the prices are not agreed, or p is not calibrated
Which model, with no cost information?	F_1, AP	used to *set* a threshold instead of compare models

Accuracy is absent from this table on purpose: report it only next to the prevalence, never as the headline for a rare event.

The ML Perspective: What Generalises Beyond Roads

A classifier is two things. A *scorer* $x \mapsto p$ learned from data, and a *decision rule* $p \geq \tau$ chosen from costs. Training touches the first; the second is set afterwards and can be changed without retraining.

Every metric is a function of four counts. Fix τ, count TP, FP, FN, TN; accuracy, precision, recall, F_1, cost are all arithmetic on them. Change τ and all of them move together.

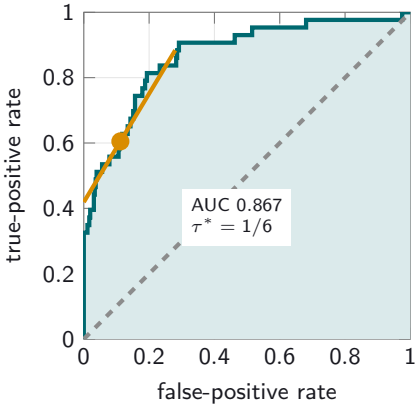
Ranking and probability are separate qualities. ROC and AUC judge the order the scorer puts examples in; calibration judges whether its numbers are frequencies. A model can have either without the other.

Decisions are risk minimisation. With prices $c_{\mathrm{FP}}, c_{\mathrm{FN}}$ and a calibrated p, choosing the cheaper action for each example gives $\tau^* = c_{\mathrm{FP}}/(c_{\mathrm{FP}} + c_{\mathrm{FN}})$ and a floor, the Bayes risk, that no threshold can beat. Perfect classification needs the features to determine the label; overlap in the scores is the visible sign that they do not.

Imbalance is a price in disguise. With rare positives, equal prices make the risk-minimising rule ignore them; reweighting, oversampling and threshold moving are all ways of saying a miss costs more.

All of this was measured on training data. Whether the curves and the threshold survive new examples is the next lecture.

Summary



1. Model + threshold = confusion matrix; every scalar metric is a function of its four cells.
2. Accuracy is owned by the majority class; at $\pi = 0.1075$ it cannot see 28 misses.
3. ROC/AUC describe the *ranking* over all thresholds; $\mathrm{AUC} = \Pr(s_+ > s_-)$.
4. PR divides by closures, so it feels a rare class; ROC does not. Calibration asks whether p is a frequency; neither curve can tell.
5. Prices give $\tau^* = c_{\mathrm{FP}}/(c_{\mathrm{FP}} + c_{\mathrm{FN}})$; iso-cost lines on the ROC plane have slope $c_{\mathrm{FP}}\, n_- /(c_{\mathrm{FN}}\, n_+)$.

Next. Every number here uses training data. We must test whether the conclusions survive on new road segments.

Before Friday's Tutorial

The tutorial is on laptops, NumPy only, and is mostly about the *model* behind today's numbers. You will receive the 400 segments and:

- build the logistic loss, its gradient $X^\top(\mathbf{p} - \mathbf{y})$ and its Hessian $X^\top W X$, and check the gradient numerically;
- fit by gradient descent and by Newton, and reproduce the probabilities on these slides to three decimals;
- add an L_2 penalty, trace the shrinkage path, then break the fit with a near-duplicate feature and repair it with Ridge;
- finish with the confusion matrix, AUC by pair-counting, and the cost at $\tau = 0.5$ against $\tau = 1/6$, all on your own $\widehat{\mathbf{p}}$.

Bring

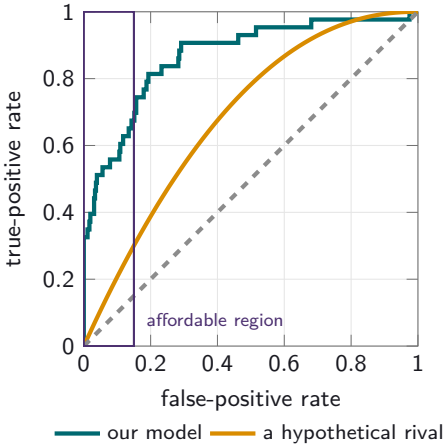
A laptop with Python and NumPy. The starter file and the data are on the portal under Tutorial 4.

Open the metrics and thresholds explorer ↗

Appendix: if time

Seven slides that sharpen the main path: what AUC ignores, ROC against PR at one threshold, ranking against probability values, why the empirical minimum drifts from τ^*, iso-cost lines, and the price-ratio curve.

What AUC Does Not Tell You



The orange curve has a *similar* area but is worse below $\mathrm{FPR} \approx 0.15$, the only region a road authority can afford.

Warning

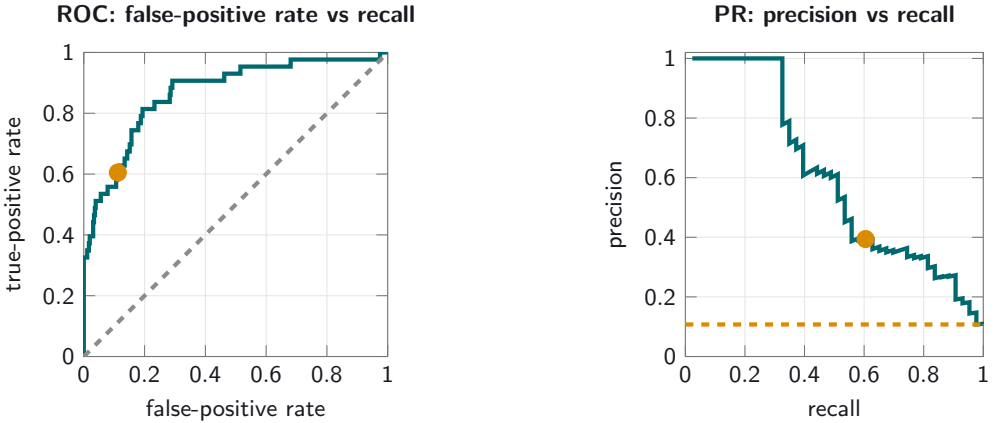
Three things AUC ignores:

1. **the threshold**: it averages over all of them, absurd ones included;
2. **the prevalence**: within-class rates look the same at $\pi = 0.1$ and 0.5;
3. **the costs**: a miss and a false alarm weigh the same.

Insight

Use AUC to compare *rankings*. Once you know where you will operate, compare the curves *there*.

ROC And PR Side By Side At One Threshold



At $\tau = 1/6$: $(\mathrm{TP}, \mathrm{FP}, \mathrm{FN}, \mathrm{TN}) = (26, 40, 17, 317)$, so ROC shows $(0.112, 0.605)$ while PR shows $(0.605, 0.394)$.

Conclusion. Both points come from the same confusion matrix. PR exposes the 40 unnecessary closures because precision divides by all closures.

AUC Sees Order; Calibration Sees Probability Values

Consider four roads. The second model changes every score but keeps the order.

	road A	road B	road C	road D
actual outcome	impassable	passable	impassable	passable
model 1 score	0.90	0.70	0.60	0.20
model 2 score	0.99	0.85	0.69	0.06
rank in both models	1st	2nd	3rd	4th

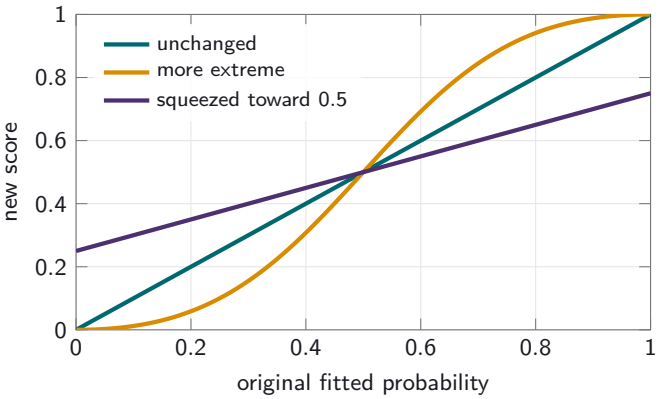
Ranking question

For each impassable–passable pair, which road receives the higher score? Both models get the same 3 of 4 pairs right, so both have AUC $= 0.75$.

Probability question

Does a score such as 0.70 correspond to about a 70% event rate? The two models cannot both be calibrated because their probability values differ.

Discrimination And Calibration Are Different Questions



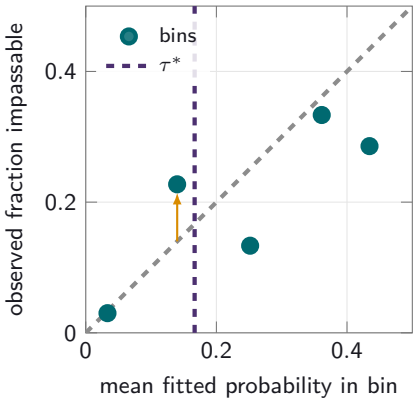
All three curves rise from left to right. Therefore they preserve the ordering of every pair and have the same ROC curve and AUC.

- **Discrimination**: who is ranked riskier? Measure with ROC/AUC.
- **Calibration**: what does the number p mean? Check with a reliability diagram.

Warning

The cost threshold on the next slides treats p as a real probability. Good ranking alone is insufficient for that decision.

Why The Empirical Minimum Drifts From τ^*



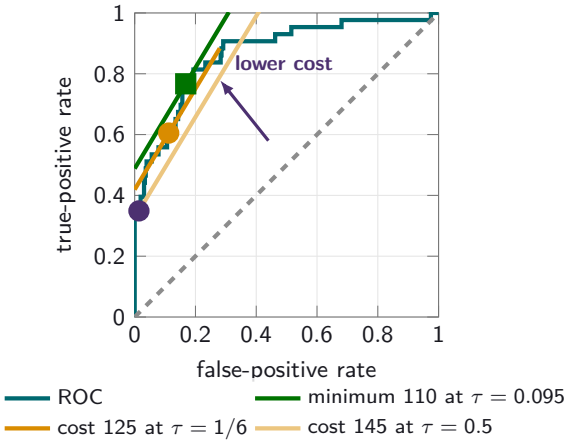
The derivation assumed p is the true probability. In the bin around $p \approx 0.14$ the observed rate is 0.227: the model *under*-states the risk there, so the true probability crosses $1/6$ at a fitted value below $1/6$.

- Miscalibration shifts the best empirical threshold.
- With 43 positives the cost curve is also noisy: bins of 7 or 15 segments cannot pin a rate to better than ± 0.1.

Warning

Do not tune τ to the minimum of a cost curve drawn on the *training* set. That minimum is partly noise; whether it survives on new data is the subject of the next lecture.

Iso-Cost Lines: Slide Up-Left Until The Line Touches



Problem. For $c_{\mathrm{FP}} = 1$ and $c_{\mathrm{FN}} = 5$, which ROC point has the lowest observed cost?

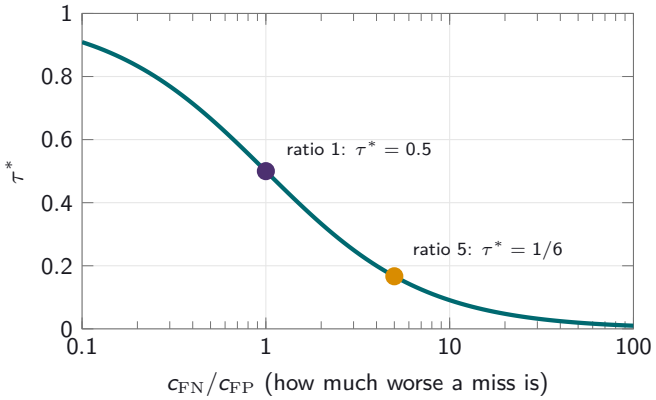
1. Draw a line of slope 1.66 through any point on the ROC curve.
2. Slide it parallel toward the **upper-left**; cost decreases.
3. The last line still touching the curve selects the threshold. Here: $\tau = 0.095$, cost 110.

Why a straight line. Cost $= c_{\mathrm{FP}} n_- \,\mathrm{FPR} + c_{\mathrm{FN}} n_+ (1 - \mathrm{TPR})$ is linear in the two rates, so equal-cost points form a line of slope $c_{\mathrm{FP}} n_- / (c_{\mathrm{FN}} n_+) = 357/215 = 1.66$.

Theory versus this sample. The calibrated-probability rule $\tau^* = 1/6$ costs 125 here. Imperfect calibration and finite-sample noise move the observed minimum, as the previous slide showed.

Interactive explorer: turn on the iso-cost line, then change the two prices ↗

The Threshold Is A Function Of The Price Ratio



$$\tau^* = \frac{c_{\mathrm{FP}}}{c_{\mathrm{FP}} + c_{\mathrm{FN}}} = \frac{1}{1 + c_{\mathrm{FN}}/c_{\mathrm{FP}}}.$$

Only the *ratio* matters; the currency does not.

setting	τ^*
a miss is as bad as a false alarm	0.50
a miss is 5× worse (today)	0.17
a miss is 20× worse (a school bus route)	0.048
a false alarm is 3× worse (hospital access road)	0.75

Warning

The costs are an engineering input, argued with the road authority before any model is trained. Choosing $\tau = 0.5$ by default is choosing $c_{\mathrm{FN}} = c_{\mathrm{FP}}$ without saying so.